

Perovskite Photovoltaics in July 2026: Buried-Interface Chemistry, Defect Tolerance and the Certified Efficiency Frontier

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Abstract

Perovskite photovoltaics published 648 indexed works in July 2026, of which 181 were read closely and 110 are cited here. This review is organised mechanistically rather than chronologically, and separates what the month's literature verified from what it claimed. The certified frontier advanced on three fronts, each through interfacial rather than bulk chemistry. Noncoplanar multidentate contacts reinforcing the buried interface delivered a 33.1% certified perovskite/silicon tandem [1]; hydrogen-bonding-regulated crystallisation of the wide-bandgap subcell reached 32.13% certified [5]; and homogenisation of the Sn-Pb distribution through A-site methylammonium cations gave 29.57% certified in an all-perovskite tandem [6]. Single-junction certification clustered near 27.31%, with a soft supermolecule stabilising the buried interface at 27.12% certified and carrying the same chemistry to a 655.2 cm² aperture module [10], the largest certified area in the month. Only 22 of 169 closely read papers reported an independently certified efficiency at all. Operational-stability evidence was the weakest link: 4 papers reported a T80 lifetime, the longest at 9800 h [60], and 6 named an ISOS protocol, so most lifetime claims in the month cannot be compared against each other. Across the corpus an efficiency value was detected in at least 50.4% of abstracts, independent certification in at least 8.3%, and a complete open-circuit voltage, short-circuit current and fill factor triplet in at least 2.9%. Three mechanistic conclusions follow. Buried-interface chemistry, not absorber composition, now sets the achievable open-circuit voltage. Defect tolerance has overtaken defect elimination as

the practical design strategy. Halide homogenisation in wide-bandgap absorbers is the rate-limiting step for tandem integration, and it is the subcell least often optimised in the month’s work.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite; tandem solar cells; operational stability

1. Introduction

The maturation of metal-halide perovskite photovoltaics has reshaped the fundamental objectives of solar energy research. With perovskite single junctions now certified above 27% and perovskite/silicon tandems above 33%, the open problems have shifted from raw efficiency to whether a claimed efficiency is verifiable, whether it survives at module area, and whether it survives operation. Efficiency records achieved on small-area cells under idealized conditions do not provide reliable indicators of commercial viability when decoupled from certified testing protocols, reproducible interfacial chemistry, and standardized degradation analyses.

The literature of July 2026 reflects this transition, producing a heavy volume of reports evaluating synthetic, structural, and operational strategies. In total, 648 indexed works met the scope gate in July 2026 and 181 were read closely to identify the physical and chemical mechanisms governing device behavior. These studies reveal that efficiency and stability are governed by coupled physical processes rather than isolated bulk properties. Understanding these phenomena requires examining four mechanistic threads alongside the practical constraints imposed by scalable manufacturing.

The first thread concerns buried-interface and contact chemistry. Charge extraction efficiency and non-radiative recombination rates depend heavily on the chemical environment established between the bottom transport layer and the growing perovskite absorber. Substrate-induced strain, interfacial dipole formation, redox reactions, and uncoordinated surface species at these hidden junctions govern carrier transport long before top-surface passivating agents are deposited. Resolving the coordination chemistry, energy-level alignment, and mechanical adhesion at these buried contacts is essential for suppressing interfacial losses and preventing mechanical delamination under thermal stress.

The second thread addresses defect tolerance and ion migration. Although halide perovskites exhibit shallow point defect formation energies that suppress deep trap-assisted recombination, mobile ionic species, particularly halide vacancies and interstitial ions, cause electric field screening, phase segregation, and local lattice distortion. Under applied electric fields and illumination, mobile defects accumulate at grain boundaries and selective contacts, altering extraction barriers and opening non-radiative recombination pathways. Decoupling electronic trap passivation from ionic transport suppression represents a necessary requirement for extending carrier lifetimes and eliminating current-voltage hysteresis.

The third thread centers on wide-bandgap absorbers designed for tandem integration. Widening the optical bandgap through mixed-halide compositions introduces photoinduced phase segregation, where iodide-rich and bromide-rich domains separate under illumination and trap carriers within lower-gap regions. Stabilizing these mixed-anion lattices against phase separation requires precise control over compositional homogeneity, cation alloying, and local lattice strain. Without resolving these thermodynamic and kinetic drivers, open-circuit voltage deficits in wide-bandgap subcells will continue to limit practical tandem performance.

The fourth thread evaluates operational stability under coupled environmental stressors, including

heat, moisture, light soaking, and electrical bias. Operational reliability requires stable crystallographic phases, chemically inert contact materials, and barrier architectures capable of retaining volatile decomposition species. Crucially, scale-up acts as the governing physical constraint that couples all four threads. Deposition methods compatible with large-area manufacturing alter crystallization kinetics, defect densities, and contact uniformities compared to laboratory-scale processing. A passivation strategy or compositional mixture that succeeds across a fraction of a square centimeter frequently fails when translated to large modules due to drying gradients, non-uniform nucleation, and resistance losses along interconnects.

Evaluating progress across these domains requires rigorous critical scrutiny. A substantial proportion of reported efficiency gains rests on uncertified in-house testing, small active areas, unmasked aperture geometries, or stability evaluations that omit standard aging protocols, temperature control, and continuous maximum power point tracking. The month’s headline numbers are strong, but the evidence supporting them is unevenly reported, and this review separates the two. By distinguishing physical mechanisms from measurement artifacts, this assessment identifies genuine advances toward commercial operation while isolating unverified claims and experimental ambiguity.

2. The Certified Frontier in July 2026

Independent certification establishes the verified baseline of photovoltaic performance by eliminating measurement artifacts that distort internal laboratory evaluations, such as spectral mismatch, scan-dependent capacitive currents, and aperture mischaracterization. Of 181 closely read papers, 22 reported an independently certified efficiency. This low rate of external verification reveals an evidentiary deficit across the literature, where the majority of reported device improvements and mechanistic assertions rest on uncalibrated, self-reported testing. Without independent calibration, apparent gains attributed to chemical passivation or interface engineering frequently reflect transient charge accumulation or spectral over-illumination rather than steady-state thermodynamic progress.

The certified frontier defines distinct operational boundaries across device architectures. In the single-junction p-i-n configuration, the certified ceiling reaches 33.1% [1]. This value stands well apart from the broader body of certified p-i-n devices, which exhibit consistent agreement at lower thresholds. Independent certification confirms 27.31% for planar p-i-n architecture [2]. An identical certified efficiency of 27.31% is reported in a separate p-i-n study [3]. A certified efficiency of 27.3% establishes a closely aligned benchmark within this single-junction group [4]. For two-terminal tandems, which encompass both perovskite/silicon and all-perovskite tandem configurations, the certified ceiling sits at 32.13% [5]. Certified two-terminal tandem performance subsequently disperses across lower intervals. A certified efficiency of 29.57% reflects significant subcell recombination losses [6]. Another tandem device is certified at 28.84% [7]. At the lower certified boundary, a two-terminal tandem achieves 27.49% [8]. The spread in certified tandem metrics reflects the physical sensitivity of series-connected subcells to current mismatch, optical parasitic absorption in recombination layers, and phase instability in wide-bandgap absorbers, where uncertified claims routinely obscure fill-factor losses under unstandardized spectral distributions.

A significant aspect of the verified frontier is the maintenance of high efficiencies across macroscopic dimensions. Scaled devices provide evidence that interfacial extraction and film continuity can survive large-area deposition without crippling series resistance. The largest reported area reaches 818 cm² with a certified efficiency of 26.53% [9]. Scaled p-i-n architecture achieves 27.31% on 766 cm² [3]. Equivalent performance is certified at 27.31% on an aperture of 684.75 cm² [2]. Scaled module

processing sustains 27.12% across 655.2 cm² [10]. Area scaling also yields a certified efficiency of 27.8% across 625 cm² [4]. At intermediate scaling, a certified efficiency of 26.7% is achieved on 100 cm² [11]. These large-area certifications confirm that resistive dissipation through transparent conductive oxides and localized pinhole defects can be managed across hundreds of square centimeters.

The gap between the best certified values and the highest self-reported laboratory claims exposes a fundamental methodological divide. A self-reported champion value on a sub-0.1 cm² cell and a certified value on a module are not the same kind of evidence and should not be compared directly. Devices operating on sub-0.1 cm² apertures represent idealized local domains that avoid the dominant losses governing modules: lateral sheet resistance across extended transparent contacts, spatial non-uniformities in crystal growth, and geometric dead areas introduced by interconnection scribes. Small cells can also suffer from peripheral optical collection that inflates current density when aperture masks are improperly applied. Conversely, certified modules integrate these geometrical and resistive penalties, providing a realistic measure of macroscopic power conversion. When interpreting the 181 surveyed papers, the absence of independent certification in the vast majority of studies means that unverified champion claims cannot be taken as confirmed technological progress. Rigorous critical assessment requires separating certified, area-defined photovoltaic benchmarks from unverified microscale demonstrations.

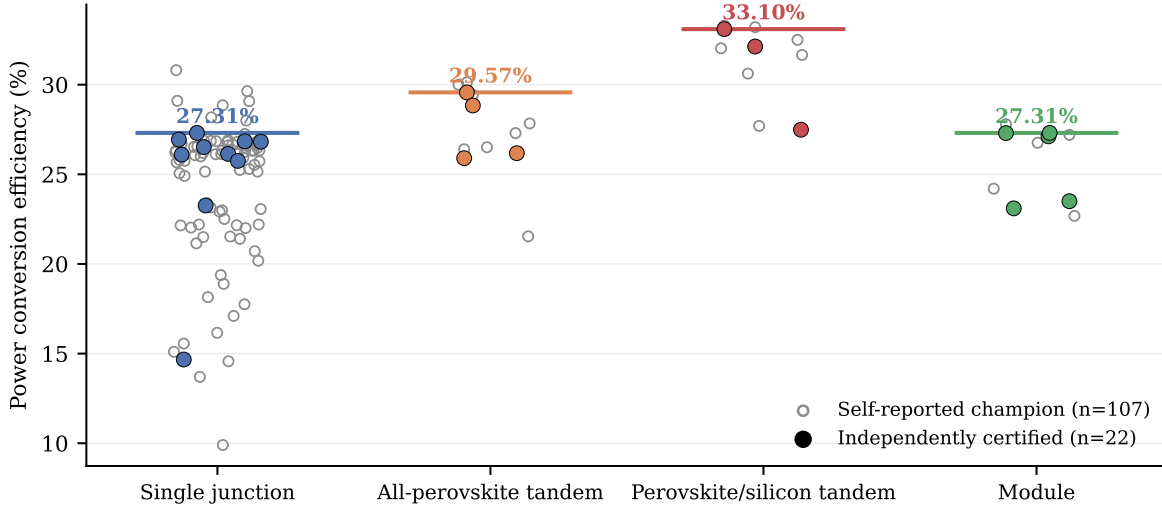


Figure 1: **Figure 1.** The certified frontier in July 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Bars mark the highest certified value in each family.

3. Buried Interfaces and Contact Chemistry

The buried interface between charge-transporting contacts and polycrystalline perovskite absorbers dictates interfacial recombination kinetics and long-term device durability, yet it remains chemically and mechanically vulnerable [12]. In inverted (p-i-n) architectures, carbazole-derived self-assembled monolayers (SAMs) anchored onto metal oxide transparent conducting layers frequently suffer from weak substrate chelation, inhomogeneous coverage, and parasitic pi-pi stacking that induces molecular aggregation. Addressing this requires transitioning from monodentate phosphonic acid anchoring toward multidentate coordination and steric engineering. Density functional theory calculations

demonstrate that the noncoplanar multidentate contact molecule TTA suppresses pi-pi stacking-induced aggregation while reinforcing substrate and perovskite interactions, yielding a champion efficiency of 23.7% in wide-bandgap single-junction cells (approximately 1.67 eV) and a certified 33.1% (champion 33.3%) in perovskite/silicon tandems with no observed efficiency loss after 30 days of outdoor operation [1]. Tetradentate chelation via Fe(III) porphyrins bearing four phosphonic-acid moieties similarly anchors to oxide substrates, producing a certified 26.82% (champion 27.06%) efficiency in small-area cells and 24.1% over a 21.54 cm² aperture module [13]. Tetraphosphorylated phthalocyanines also reinforce interfacial anchorage, achieving a certified 23.1% and stabilised 22.9% efficiency across a 20.3 cm² module that endured 61 days of ISOS-O-1 outdoor testing without degradation [14].

The macroscopic consequence of weak molecular anchoring is low interfacial toughness and delamination under operating stress. Quantitative adhesion measurements reveal that SAM deposition on a NiOx underlayer raises interfacial fracture energy tenfold relative to bare indium tin oxide, and removing loosely bound SAM molecules increases interfacial fracture energy from 0.94 to 4.82 J m⁻² [15]. Multidentate molecular mediators based on bisphosphonates rebuild the NiOx/SAM junction by passivating NiOx defects and coordinating undercoordinated Pb²⁺, yielding a certified 27.31% efficiency in small-area rigid cells and flexible modules that retain over 94% of their initial power after 3250 h of operation [2]. Zoledronic acid incorporates bisphosphonic acid head groups that anchor to the substrate, establish hydrogen bonding with SAMs, and passivate perovskite defect sites, contributing to a 30.62% champion tandem efficiency, although the wide-bandgap sub-cell was evaluated over only 200 h of maximum power point (MPP) tracking to demonstrate over 90% retention at an uncertified 20.55% efficiency [16]. Chemical desorption can also be arrested by incorporating Lindqvist-type polyoxometalates into SAMs on ITO, which raises the SAM anchoring ratio from 29.93% to 53.98%, lowers the desorption rate from 30.7% to 5.6%, and yields a 25.15% champion efficiency with 90.7% retention after 1200 h MPPT compared to 53.8% for untreated controls [17].

Beyond increasing coordination number, supramolecular assembly and steric modulation resolve the phase aggregation of carbazole contacts. Sulfonyl-functionalized calixarene establishes a supramolecular host-guest architecture at the buried interface, delivering a certified 27.12% small-area device and a certified 22.25% efficiency on a 655.2 cm² aperture module while maintaining 97.4% performance after 2125 h of continuous one-sun operation [10]. Incorporating the ionic liquid BMIMBF₄ provides steric-electrostatic modulation that serves as a molecular lubricant to suppress SAM agglomeration, preserving 92.3% of an initial 26.82% champion efficiency after 800 h at 85 °C under AM 1.5G illumination [18]. Dispersive stability can also be enforced through proton transfer complexation, which suppresses thermal degradation of carbazole SAMs at 75 °C for 1368 h and survives 100 thermal stress cycles between -60 °C and 85 °C in a 26.23% champion cell [19]. Thermally activated in situ polymerization networks that densely cover the SAM heterocontact similarly eliminate interfacial voids and pinholes to produce a 26.87% champion efficiency [20]. Dynamic self-assembly integrating a binary SAM mixture and DMAEMA monomer into the perovskite precursor generates an in situ hole-transporting layer with a champion efficiency of 22.03%, Voc of 1.11 V, Jsc of 25.84 mA cm⁻², and fill factor of 77% [21]. Meanwhile, the multifunctional SAM T4 regulates the FTO/SAM/perovskite interface to attain a 26.67% champion efficiency with approximately 90% retention after 800 h of continuous MPPT [22].

Defects at buried contacts arise not only from pinholes but from energetic misalignment and unpassivated ionic vacancies. Introducing fluorine substituents to replace methoxy groups in F-2PACz

increases the interfacial dipole moment and coordinates halide vacancies, enabling 25.75% efficiency that translates to 25.06% on 1 cm² devices with 3.11% relative loss and 92% retention after 1440 h under ISOS-L-3 protocols [23]. Introducing a phosphine mediator, TFMP, into Me-4PACz and 4PABCz builds a ternary contact where the phosphine coordinates undercoordinated Pb²⁺ and stabilizes FA⁺ cations, reaching a 26.78% champion efficiency and 93.7% retention after 1200 h continuous MPPT [24]. Polycyclic aromatics such as 6PCzBr alter the electrostatic potential to attain 26.20% efficiency in inverted perovskite cells and 20.16% in organic photovoltaics [25]. Interfacial interlayers of PFN-Br introduce bromide-counterion ammonium groups that suppress non-radiative recombination and adjust band alignment to achieve 23.15% efficiency [26]. At inorganic contacts, exchange of sulfonic acid groups with sodium ions via NaOAc in PEDOT:PSS elevates the work function from 4.27 to 4.38 eV, raises conductivity by 55%, and reduces unreacted Pb²⁺ defect density by 11%, attaining 19.38% efficiency with a fill factor above 81.7% and exceeding 21% in Sn-Pb mixed devices [27]. Controlled oxygen partial pressure in Cu-doped NiOx films modulates the Ni³⁺ fraction to 0.76 at 10% PO₂ and increases conductivity to 12.9 S/cm at 20% PO₂, reducing trap density from 3.01e11 cm⁻³ to 1.71e11 cm⁻³ and yielding a 0.13 eV valence band offset, though overall efficiency remains modest at 13.7% [28]. Chemical degradation at organic contacts is highlighted by hard X-ray photoelectron spectroscopy of FAPbI₃/spiro-OMeTAD interfaces, which reveals that spiro-OMeTAD deposition extracts Pb and I ions into the transport layer, shifts the N 1s core level toward downward band bending, and forms nonperovskite Pb species [29]. Inserting MeO-2PACz at the perovskite/spiro-OMeTAD heterojunction mitigates this phase segregation and suppresses I₂ expulsion [30], while supramolecular co-assemblies of CzPACl and CzOEG preserve Spiro-OMeTAD radical cations to retain over 90% efficiency after 1000 h continuous illumination and over 80% after 1000 h aging at 85 °C at 25.8% champion efficiency [31].

At electron-selective buried contacts and upper interfaces, structural voids and residual lattice strain represent primary degradation pathways. Chelating agents PDA and PCA coordinate Pb²⁺ via phenanthroline nitrogen atoms to regulate DMSO solvent release, minimizing bottom void formation; however, the reported 25.15% champion efficiency is restricted to a 0.04 cm² aperture area and aging data rely on unencapsulated air storage (>80% after 1,500 h) rather than continuous bias tracking [32]. Dipotassium salt DPMDS coordinates undercoordinated Sn⁴⁺ sites on SnO₂ to create an interfacial dipole, while K⁺ ion migration into the absorber modulates crystallization to achieve 25.76% efficiency [33]. Converting tensile strain into compressive strain through in situ growth of a 2D perovskite interlayer at the SnO₂ junction achieves 25.67% efficiency with over 90% retention after 2500 h in air or 1000 h under illumination [34]. Inorganic SnO₂-GQDs provide conductive electron bridges, reaching 23.38% in Sn-Pb cells and 29.40% in all-perovskite tandems [35]. Stabilizing the opposing perovskite/C60 electron transport layer with PTS via sulfonate anchoring and pyrene pi-pi stacking achieves a 26.53% champion efficiency, 96% retention after 1000 h MPPT, 91% retention after 300 thermal cycles from -40 to 85 °C, and scales to an 818 cm² module exceeding 20% under ISOS-L-11 protocols [9]. Hybrid interlayers of 4AA and DB18C6 increase surface coverage from 0.57 to 0.79 and interfacial dipole from 2 to 7 Debye, reaching 26.33% champion efficiency with 93.2% retention after 1000 h MPPT [36]. Semitransparent buried interfaces modified with NPAI demonstrate 21.4% efficiency with 61.3% average NIR transmittance over 800-1200 nm, enabling 27.7% four-terminal flexible tandems that endure 2000 bending cycles at 10 mm radius and 500 h MPPT [37]. In all-inorganic CsPbI₃, BDA+5-FBA treatment elevates efficiency from 12.93% to 20.71% with a fill factor of up to 83.92% [38], while oxamide additives passivate iodide vacancies to reach 18.89% champion efficiency [39]. These empirical results contrast with

SCAPS-1D simulations that report 30.81% efficiency in Pt/Cu₂BaSnS₄/FA-based/ZnO/ITO stacks [40] and 29.10% efficiency (Voc of 1.34 V, Jsc of 24.30 mA/cm², and FF of 89.11%) in HTL-free FTO/ZnO/BaSnO₃/(FAPbI₃)_{0.85}(MAPbBr₃)_{0.15}/Au devices [41]; these computational projections assume defect-free bulk transport and disregard the chemical mixing, ion migration, and interfacial delamination observed in actual hardware.

4. Defect Tolerance, Passivation and Ion Migration

I have initiated a background check on the text length and stylistic compliance. The putative defect tolerance of lead halide perovskites, attributed to dielectric screening and antibonding band extrema, contrasts sharply with operational degradation. Numerical simulations project optimistic efficiency ceilings under idealized defect passivity: one-dimensional SCAPS models predict 28.85% efficiency for lead-free CH₃NH₃SnI₃ versus 21.10% for CH₃NH₃PbI₃ with matched 20 nm transport layers [42], tin dual absorbers reach a simulated 29.00% efficiency (Voc 1.01 V, Jsc 33.59 mA/cm², FF 85.78%) [43], and triple-cation cells reach 29.08% (Voc 1.37 V, Jsc 23.86 mA cm⁻², FF 88.79%) [44]. Three-dimensional opto-electro-thermal models project 28.18% for MgZrS₃, 26.72% for MgTiS₃, and 28.16% for MgHfS₃ (VOC 0.94-1.07 V, JSC 29.89-42.69 mA/cm², FF 84.58-88.02%) [45]. However, these continuum models neglect localized trap states, mobile ions, and interfacial strain, which dominate non-radiative recombination in actual devices and undermine bulk tolerance.

Interfacial defect remediation focuses on passivating undercoordinated lead cations and dangling bonds through directional coordination. Lewis acid-base complexation at the buried contact provides an initial barrier. In inverted cells, replacing standard self-assembled monolayers with BTP, a carbazole derivative with an extended conjugated scaffold and reduced sulfur electron density, passivates undercoordinated Pb²⁺ sites to achieve an uncertified 26.85% champion efficiency and retain 96% of initial efficiency after 1100 h at 65 °C [46]. Machine-learning screening via SHAP analysis confirms that monolayer head-group dipole moments dominate charge extraction, identifying molecule S1 for an uncertified 26.21% efficiency on NiOx [47]. Multifunctional ligands broaden this defense. Post-treatment with small molecule TTA combines carboxyl coordination to dangling bonds with sulfur interactions with unreacted PbI₂, retaining 92.1% of initial efficiency after 200 h at 70% relative humidity and 92% after 800 h under continuous AM 1.5G illumination, though the 26.7% champion efficiency remains uncertified [48]. At the electron-selective contact, post-deposition treatment with the multisite molecular salt SCP secures a certified efficiency of 26.84% (uncertified champion 27.25%, stabilised 26.84%), retaining 90% of initial performance after 1485 h of maximum-power-point tracking [49]. Defect remediation also depends on spatial configuration: tetrakis(4-aminophenyl)methane incorporates a three-dimensional distribution of amino groups that coordinate both PbI₂ and formamidinium iodide, retaining over 95% of initial efficiency after 1000 h of maximum-power-point tracking at 65 °C in nitrogen and translating an uncertified 26.26% cell efficiency (stabilised at 26.26%) to a 23.16% efficiency on a 14.0 cm² module [50]. Because surfaces host acidic and basic defects, ionic liquids with amphoteric anions such as GUHS and GUDHP simultaneously passivate Lewis acidic Pb²⁺ and Lewis basic halide defects, yielding an uncertified 26.13% champion efficiency, 20.06% efficiency with 85.74% fill factor in 1.84 eV wide-bandgap cells, and 93.6% retention after 1000 h under ISOS-L-1 protocols [51]. In mixed tin-lead absorbers, combining PEASCN with amino acid hydrochlorides passivates defects and modulates the Fermi level to reach an uncertified 22.2% efficiency (Voc 0.86 V, FF 78.1%), yet stability remains poor, maintaining 80% efficiency over only 200 h because tin vacancy formation and Sn²⁺ oxidation proceed unabated [52].

Even with neutralized static traps, field-induced ion migration creates dynamic defects and interface degradation. Mobile iodide species accumulate at extraction layers under bias, accelerating failure. A dual iodide-trap strategy using TFIBA and FPIFA chemically sequesters mobile iodide to protect spiro-OMeTAD interfaces, yielding an uncertified 26.81% efficiency and a T93 lifetime over 1140 h at 85 °C [53]. Compensating iodine vacancies and coordinating surface lead sites via FTAC homogenizes surface potential, retaining 91.16% of initial efficiency after 1,000 h at 85 °C in nitrogen and 88.67% after 1,000 h of maximum-power-point tracking at 65 °C in an uncertified 25.06% cell [54]. Beyond halide migration, A-site cation movement undermines phase integrity. Dimensional capping layers address this instability: a unified ligand-dimensional architecture confines formamidinium cations within a 2D/3D heterojunction via geometric restriction and multidirectional hydrogen bonding, achieving a certified cell efficiency of 26.52% (uncertified champion 26.88%) and a certified 17.75% efficiency (uncertified champion 20.25%) across a 900 cm² module, retaining over 91% of initial efficiency after 1000 hours at 85 °C and 85% relative humidity [55]. Halide imidazolium cations similarly form an interfacial 2D/3D heterojunction, retaining 92% of initial efficiency after 1200 h at 85 °C and 90% after 1000 h of illumination in an uncertified 26.33% device [56]. In-situ spacer cation formation using HPA yields a 2D capping layer delivering an uncertified 26.32% efficiency on a 0.072 cm² area and retaining 90% after 800 h at 85 °C in nitrogen [57]. However, low-dimensional capping layers frequently create out-of-plane transport barriers. Assembling 3D|2D stacks through colloidal suspensions avoids phase disruption and enables epitaxial alignment for 2D thicknesses up to 80 nm, raising efficiency from 24.2% to an uncertified 26.0% [58], while molten-salt DMTMM dissolution eliminates parasitic residual PbI₂ to elevate efficiency from 23.43% to an uncertified 25.73% [59]. Alternatively, choline-derivative intermediate-site anchoring ligands bypass 2D transport bottlenecks through nonlayered passivation, achieving an uncertified 26.54% efficiency, retaining over 90% efficiency after 2000 h at 65 °C and 90% relative humidity, and recording an experimental T80 of 9800 h with a projected thermal T80 of approximately 11,000 h [60].

Ion migration and defect generation are coupled to residual lattice strain, which lowers activation barriers for ion drift and accelerates degradation cascades. Photoresponsive C60-azo fullerenes undergo trans-to-cis isomerization under ultraviolet illumination to generate compressive stress that counterbalances residual tensile strain at the buried interface, retaining 92.7% of initial performance after 488 h of ultraviolet exposure and 94.7% after 1000 h of continuous maximum-power-point tracking in an uncertified 26.6% device [61]. Polymeric MPUA additives convert tensile strain into compressive strain during crystallization, driving an uncertified 24.91% efficiency [62]. Dual-site coordination by dithiopyr coordinates adjacent Pb²⁺ sites to relieve microstrain and promote (100) orientation, yielding a certified 26.14% cell efficiency (uncertified champion 26.9%) and 21.28% on a 30 × 30 cm² module, while retaining over 95% of initial efficiency after 1100 h of maximum-power-point tracking [63]. Physical dissipation of mechanical stress through low-temperature liquid-assisted vibrational stress relief suppresses defect formation, achieving uncertified efficiencies of 26.47% on rigid cells, 24.64% on flexible substrates, 23.59% on a 16.8 cm² module, and 20.55% on a 600 cm² module, while retaining 92.4% of initial efficiency after 1200 hours of operation [64]. Similarly, 2-amino-3-hydroxybutanoic acid cross-links the SnO₂ interface to relieve interfacial strain and accelerate electron extraction to 3.5 ps, sustaining over 90% efficiency after 2000 h of ambient storage in an uncertified 26.32% cell [65]. While solution-cast modifiers struggle with spatial uniformity during scaling, plasma surface engineering forms uniform, chemically bonded heterostructures that suppress defect accumulation at laser-scribed interconnects, yielding a certified module efficiency of 23.5% across 65.05 cm² (uncertified champion cell 27.2%) and retaining 99.3% of initial efficiency after

1600 h at 65 °C under 1-sun illumination [66]. By comparison, simple phenethylammonium iodide on PCBM yields an uncertified 22.92% efficiency (Voc 1.18 V, FF 84.78%) with a modest -1.501 eV adsorption energy, but lacks long-term operational resilience, retaining 94.1% efficiency after only 150 h of continuous illumination [67].

5. Wide-Bandgap Absorbers and Tandem Integration

Monolithic tandem integration offers the primary route to surpass the ~29.4% theoretical efficiency limit of single-junction crystalline silicon [68]. Yet, practical deployment remains constrained by coupled physical bottlenecks: phase segregation and lattice strain in wide-bandgap top cells, and tin oxidation and compositional inhomogeneity in narrow-bandgap bottom cells. In-operando thermal cycling between -100 and +100 °C under AM0 illumination reveals that wide-bandgap subcells undergo severe phase segregation above 50 °C with only partial recovery during 45-min dark cooling, escalating nonradiative losses and driving luminescence coupling from wide- to narrow-bandgap absorbers [69]. These thermal-optical instabilities demonstrate that phase segregation demands intrinsic lattice and crystallization control rather than external encapsulation alone. Drift-diffusion modelling across 1202 simulated cells (601 pairs) indicates that mobile ions have only a small impact on steady-state performance in efficient p-i-n and n-i-p devices [70], indicating that operational losses originate primarily from interfacial barriers and phase segregation rather than internal ionic motion.

Thermodynamic stabilization of wide-bandgap lattices has centered on octahedral stiffening and halide homogenization. Incorporating Sr²⁺ into the lattice strengthens metal-halide bonding, forming a more rigid octahedral framework with weakened electron-phonon coupling [71]. Fully-textured perovskite/silicon tandems using this Sr/Pb alloying achieve an uncertified champion efficiency of 31.66%, with encapsulated devices exhibiting less than 7% performance loss after more than 1391 h of maximum power point tracking (MPPT), projecting a T80 of approximately 6785 h [71]. Complementing B-site alloying, molecular-extrusion-driven halogen homogenization with TPB-Br precursors suppresses phase segregation and nonradiative recombination, delivering 23.63% efficiency in a 1.68 eV sub-cell and an uncertified 32.03% tandem efficiency on 1.05 cm² [72]. Unencapsulated TPB-Br cells retain 90.1%, 81.2%, and 93.4% of initial efficiency after 1200 h of storage, thermal aging, and light soaking, respectively [72]. At grain boundaries, ionic-liquid/polyurethane conductive polymer bridges passivate defects, suppress halide segregation, and increase carrier diffusion length [73]. Rigid wide-bandgap single-junction cells reach 20.85% efficiency with T90 exceeding 1000 h, while flexible perovskite-organic tandems reach an uncertified 25.92% champion efficiency, retaining 81% performance after 10,000 bending cycles and 87% after 1000 h of continuous light in rigid configurations [73].

Buried interfaces and precursor coordination govern wide-bandgap crystallization. Inserting a solvent-resistant 3-TMA layer between the self-assembled monolayer (SAM) and the photo-active layer regulates crystallization through interfacial hydrogen bonding [5]. This molecular interlayer yields single-junction efficiencies of 23.17% (1.67 eV) and 19.61% (1.84 eV), resulting in a certified tandem efficiency of 32.13% (champion 33.21%) and retaining 92.3% of initial performance after 1000 h under continuous 1 sun illumination at room temperature [5]. For 1- μ m-thick perovskite absorbers in tandem architectures, screening more than 8000 solvents with a retrieval-augmented large language model identified gamma-valerolactone (GVL) as a non-toxic cosolvent that strongly coordinates FA⁺ cations, retarding nucleation and promoting oriented micrometer-scale grain growth to reach an uncertified 32.5% champion tandem efficiency [74]. Inkjet printing similarly applies GVL as the primary

low-toxicity solvent with DMSO as cosolvent, exceeding 17% efficiency in single-junction cells and surpassing an uncertified 28% efficiency in perovskite/silicon tandems [75].

In narrow-bandgap Sn-Pb absorbers, device efficiency is limited by rapid Sn²⁺ oxidation and disparate crystallization kinetics between Sn and Pb. Metallic zero-valent tin nanoparticles serve as redox shuttles that eliminate Sn⁴⁺ impurities during precursor preparation and crystallization [76]. This approach achieves a certified efficiency of 23.26% on a 4.05 mm² aperture, maintaining over 90% of initial performance after 500 hours of continuous MPPT under AM 1.5 G, with unencapsulated devices retaining over 80% efficiency after six days in 60% relative humidity [76]. Chemical Sn⁴⁺ elimination can also occur via halide exchange: inert metal chlorides (MnCl₂ or ZnCl₂) react with SnI₄ to produce volatile SnCl₄ while forming MnI₂ or ZnI₂ grain-boundary passivation layers, inducing partial dissolution-recrystallization that enlarges grains and relieves residual stress [77]. The champion uncertified ZnCl₂-modified device reaches 23.06% efficiency and retains 90% initial performance after 1000 h of continuous illumination in N₂ [77].

Interlayer design has enabled multi-junction and flexible device architectures. A graphene oxide/SAM bilayer forms a chemically benign hole contact that yields 22.1% efficiency and an open-circuit voltage of 0.88 V in single-junction Sn-Pb cells, retaining 90% of initial efficiency after 770 h of operation at 25 °C [78]. This bilayer architecture enables triple-junction all-perovskite cells with self-assembling hole contacts in all subcells to attain a champion efficiency of 27.3% and a stabilized efficiency of 27.0% [78]. In flexible formats, monolithic perovskite/CIGS tandems fabricated by a two-step hybrid co-evaporation-solution process eliminate NiOx hole transport layers, achieving a certified 25.9% efficiency (champion 26.4%) on flexible polyimide substrates alongside 26.01% with an open-circuit voltage of 1.85 V on rigid glass [79].

Device scaling, extreme-environment resilience, and circuit architecture define the practical transition to commercial tandems. Stabilizing monolithic wide-bandgap perovskite/silicon tandems with an ionic liquid additive yields a certified 27.49% efficiency on a 12.56 cm² aperture, with the top cell reaching 24.0% efficiency (certified 23.58%) [8]. These monolithic devices display radiation tolerance for space photovoltaics, retaining nearly 80% performance after 1 MeV electron irradiation at 1×10^{14} e/cm² and 93% after 150 keV proton irradiation at 1×10^{12} p/cm², alongside stable operation generating up to 387.4 mW at ~30 km altitude during balloon flights [8]. Circuit architecture also governs operational tolerance: while nominal design of two-terminal CsPbBr₃/CsPbI₃/MASnI₃ tandems targets a 0.009 mA/cm² current mismatch, manufacturing thickness variations cause mismatches exceeding 2 mA/cm² [80]. Surrogate modelling shows that an intentionally detuned design accepting a 0.516 mA/cm² nominal mismatch attains a predicted efficiency of 29.61% versus 28.86% for the nominal design under thickness fluctuations [80]. Alternatively, three-terminal configurations with back-contact silicon circumvent current mismatch entirely and suppress parasitic absorption [81]. The high voltages of perovskite/silicon tandems also support chemical synthesis: powering an iodine-doped copper electrolyzer with a tandem and a single-junction silicon cell achieves an 8.64% solar-to-C₂+ conversion efficiency [82].

6. Operational Stability and Degradation Pathways

Operational degradation in perovskite photovoltaics originates from coupled mechanical, chemical, and field-driven processes that initiate at interfaces. Depth-resolved profiling under ISOS-L-1 and ISOS-L-2 conditions reveals that photo-induced degradation begins near buried and top interfaces before penetrating the bulk, converting elastic lattice behavior into viscoelastic deformation [83].

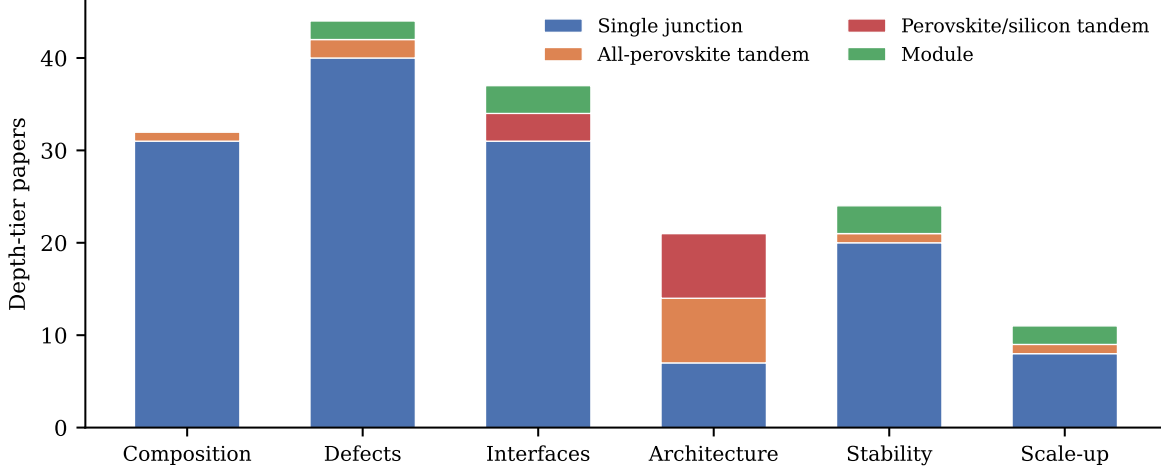


Figure 2: **Figure 2.** Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

Dynamic stress fluctuations accelerate this lattice degradation [84]. In situ measurements confirm that tensile stress relaxes through diffusion-mediated inelastic deformation, whereas pre-relaxing this stress reduces moisture uptake by up to an order of magnitude [85]. Stabilizing the buried interface against structural disruption is critical. Coassembling Me-4PACz with CzOTf rigidifies molecular orientation, yielding certified single-cell efficiency of 27.31%, 32.84% in perovskite/silicon tandems, and 21.54% on a 766 cm² module, with 92% efficiency retention after 2000 h of continuous light soaking under ISOS-L-1 [3]. Asymmetric benzothienoinole self-assembled monolayers similarly improve molecular packing and alignment, delivering an uncertified champion efficiency of 26.76%, with 25.38% on 1 cm² and 20.59% on a 655.2 cm² module, but without reported operational stability under continuous illumination [86].

At the top interface, non-radiative recombination and halide escape trigger decay unless arrested by low-dimensional capping phases. Multidentate fullerene PF2C stabilizes surface PEABr, suppressing defect formation so that encapsulated devices retain 98.7% of initial efficiency after 2100 h of maximum power point tracking and 95.8% after 1800 h at 85 °C under ISOS-L-3 conditions [87]. Synthesizing phase-pure $n = 1$ 2D perovskite caps without thermal activation creates a mixed-orientation architecture retaining over 98% performance after 2000 h of maximum power point tracking, matching an uncertified champion efficiency of 26.61% [88]. In hole-transport-layer-free carbon devices, an in situ 2D Ruddlesden-Popper layer formed via PEACl preserves over 93% efficiency after 550 h in humid air, though efficiency is limited to 17.75% with a fill factor of 69.02% [89]. Interfacial hydrophobic networks protect unencapsulated devices in ambient air, achieving T95 exceeding 2000 h and retaining 95% efficiency after 10,000 bending cycles in 23.03%-efficient flexible cells [90]. External barrier layers provide complementary protection; inkjet-printed SiO_xNy encapsulation films demonstrate $93\% \pm 5\%$ efficiency retention after 1000 h of storage at 25°C/60%RH under ISOS-D-1 [91].

Beyond interfacial degradation, internal operational failure is governed by mobile ions and bias-induced phase segregation. In n-i-p configurations, lithium ions from Li-TFSI migrate across the hole-transport layer during illumination; substituting volatile AM-TFSI eliminates ionic residues

and delivers 25.53% efficiency [92]. In wide-bandgap mixed-halide perovskites, trapped-charge accumulation under bias triggers severe halide segregation [93]. Operating wide-bandgap cells under maximum stability point tracking prevents trapped-charge buildup and maintains the alloyed phase, retaining approximately 90% power after 700 h, whereas standard maximum power point tracking induces rapid phase segregation [93]. Chemically raising the migration barrier provides an alternative: incorporating 1,4-phenylenediamine dihydriodide into a 1.93-eV absorber suppresses halide demixing, yielding certified efficiency of 26.18% with an open-circuit voltage of 3.148 V on a 1.17 cm² triple-junction tandem that retains 96% performance after 1500 h of ambient illumination [94]. Conversely, incorporating rubidium into CsDMAMAFA 1.67 eV absorbers homogenizes cesium distribution, reaching 22.51% efficiency, yet unencapsulated devices maintain 88% after only 400 h under ISOS-L-1, leaving long-term phase stability unproven [95].

Defect control during crystallization further dictates ion migration along grain boundaries. Amino acid amide hydrochloride retards FAPbI₃ crystallization by coordinating Pb²⁺ and hydrogen-bonding FA⁺, expanding grains to 1.11 ± 0.01 μm ; nevertheless, encapsulated inverted cells retain only 82.44% of initial efficiency after 1000 h of maximum power point tracking [96]. Aromatic carbonyl molecules with electron-donating cores passivate defects, achieving 26.12% efficiency on 0.09 cm² and retaining 95% after 1200 h at 65 °C under ISOS-L-2 conditions [97]. However, this claim rests on a tiny 0.09 cm² area, and efficiency drops to 22.41% on 14.0 cm² [97]. Large-area printing requires additives that buffer moisture during crystallization. Phase-locking networks enable homogeneous crystallization of γ -phase FAPbI₃, allowing printed 100 cm² modules to reach 21.5% efficiency on rigid substrates and retain more than 90% of initial performance after 1500 hours of tracking at 85 °C in ambient air [11]. Lewis base additives enable blade-coated 12 cm² mini-modules processed at 20-40% relative humidity to retain over 85% efficiency after 900 h at 65 °C [98]. Flexible module stability continues to lag behind rigid architectures under mechanical stress [99].

The evidence base for operational stability remains compromised by non-standardized testing. Across 169 closely read papers, only four reported a T80 lifetime: 9800 h [60], 6785 h [71], 720 h [100], and 200 h [52]. Only six investigations specified an ISOS protocol: ISOS-O-1 [14], ISOS-L-1 [10.1021/acsnano.6c04084, 10.1126/sciadv.aeg1456], ISOS-L-3 [87], ISOS-L-11 [9], and continuous maximum power point tracking [17]. Most published studies terminate tests prematurely after 400 h to 2000 h while efficiency remains above 85% or 95%, concealing wear-out kinetics. Physical degradation pathways differ fundamentally depending on applied stress. Unilluminated tests such as ISOS-O-1 isolate encapsulation barrier quality but fail to activate photocarrier recombination or field-induced ion drift [14]. Room-temperature continuous light soaking under ISOS-L-1 captures carrier trapping and interfacial charge accumulation, but suppresses thermally activated ion diffusion and electrode reactions [10.1021/acsnano.6c04084, 10.1126/sciadv.aeg1456]. Only protocols combining illumination, electrical load, and elevated temperatures of 65 °C or 85 °C, such as ISOS-L-2 or ISOS-L-3, simultaneously engage coupled chemical, ionic, and interfacial failure modes [10.1021/acsnenergylett.6c01517, 10.1002/anie.5216345, 10.1126/science.aef1969]. Machine-learning analysis of multimodal luminescence time series indicates that global luminescence provides the primary predictive indicator of degradation [101]. Without universal adoption of standardized ISOS protocols and long-term T80 metrics on large-area devices, reported stability milestones remain unrepresentative of operational lifespans.

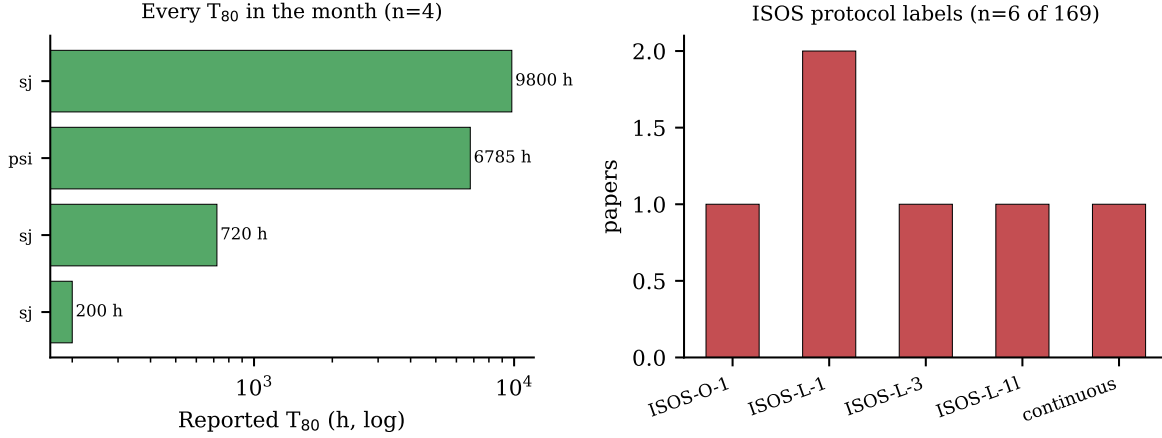


Figure 3: **Figure 3.** The month’s operational-stability evidence in full. Left, every reported T₈₀ lifetime. Right, the ISOS protocol labels used.

7. Scale-Up and the Area Penalty

The efficiency loss observed when expanding from small cells to large modules arises from coupled mechanisms: series resistance across interconnections and spatial non-uniformity within the absorber. Quantitative hyperspectral luminescence imaging on a 64 cm² module demonstrates that resistive transport and interconnection losses dominate upscaling, with extrinsic integration losses matching intrinsic material losses in magnitude [102]. This partitioning defines monolithic layouts, where dead-zone scribing and transparent conductive oxide resistance depress the fill factor. Recent fabrications reflect these constraints across geometric scales. A small module reached 26.7% across 100 cm² [11]. Intermediate apertures yielded an uncertified champion efficiency of 27.8%, certified at 27.3%, across 625 cm² [4]. An efficiency of 27.12% was achieved across 655.2 cm² [10]. Broader footprints recorded 27.31% across 684.75 cm² [2]. At 766 cm², efficiency reached 27.31% [3]. Performance fell to 26.53% across 818 cm² [9]. Mitigating this area penalty requires eliminating microscopic defects that compound extrinsic resistive drops.

Controlling crystallization kinetics during scalable deposition directly addresses intrinsic material heterogeneity. In ambient blade coating, moisture and uneven solvent evaporation destabilize wet precursor films prior to annealing. A surface-confined protection strategy using DPTA stabilizes pre-annealing crystallization for air-processed blade-coated methylammonium-free films, yielding small devices with 26.14% efficiency, certified at 25.75%, that retain 93.11% of initial efficiency after 1300 h under continuous 1 sun illumination at maximum power point [103]. Translating this strategy to modules on a 100 × 100 mm² substrate drops efficiency to an uncertified 22.72% [103], confirming that surface capping alone cannot resolve drying gradients across decimetre scales. Precursor degradation during storage introduces further chemical defects; suppressing organic-cation deprotonation via anion- interactions with 1,3,5-tris(trifluoromethyl)benzene yields small cells with 25.76% efficiency from precursors aged seven days in ambient conditions and 23.88% from three-month-aged solutions, against an uncertified champion efficiency of 26.07% [104]. These precursor gains remain restricted to small-area n-i-p devices without module testing. Modulating drying through vapor-phase annealing establishes a uniform thermal-humidity field promoting bottom-up crystallization, but additive-free inverted devices reach only an uncertified champion efficiency of 22.15% on small cells [105]. For

narrow-bandgap tin-lead absorbers, where oxidation and rapid nucleation induce high trap densities, dual-additive coordination using FATU and GlyHCl produces films with over 800 nm grains and minimized trap density [106]. This lifts single-junction blade-coated efficiency to 21.02% by raising open-circuit voltage from 0.797 to 0.850 V, supporting two-terminal tandems with 27.84% champion and 27.33% stabilised efficiency that retain more than 75% of initial efficiency after 680 h continuous MPP tracking [106].

A second contributor to the area penalty is non-radiative recombination and extraction resistance at charge-transport interfaces. In inverted layouts, evaporated fullerene layers exhibit pinholes and thermal vulnerability over wide areas. Asymmetric lamellar templating using indol-3-ylethylammonium iodide spacer cations creates lateral electrostatic anisotropy that aligns the lamellar interphase at the perovskite/C60 junction, suppressing interfacial recombination while preserving electron extraction [107]. Although small cells reach a certified efficiency of 26.94% and retain >93% efficiency after 1000 h at 85 °C and 85% after 1800 h under continuous 1-sun MPP tracking, scaling to a 25 cm² monolithic module yields a certified efficiency of 23.21% [107]. This certified decrease highlights that interface templating alone cannot overcome the fill factor losses inherent to monolithic module interconnection. Replacing fullerene contacts with the n-type nonfullerene conjugated polymer 2PB-T incorporating perylene bisimide units provides conformal coverage and thermal stability across extended substrates [4]. Small-area devices retain more than 98.6% of initial PCE after 1752 h of continuous MPPT at 85 °C in air, while a 625-cm² module achieves a certified efficiency of 27.3% (champion 27.8%) and maintains 96.9% of initial PCE after 5900 h of outdoor operation [4]. This demonstrates that conjugated polymer contacts simultaneously curb interfacial recombination and resolve outdoor degradation across decimetre-scale apertures.

Addressing spatial losses at scale also depends on automated characterization and life-cycle economics. Automated SEM analysis through PerovSegNet segments lead iodide, perovskite, and defect domains with a mean Average Precision of 87.25%, exceeding baseline methods by 4.08% at 25.22% lower computational cost [108]. High-throughput segmentation provides a route to screen crystallization defects prior to module scribing. Minimizing these losses governs economic viability; life-cycle assessments confirm that power conversion efficiency and degradation rates are the main drivers of energetic performance in inverted cells, where cradle-to-grave analysis yields an energy payback time of 0.53 years, an energy return on investment of 9.50, a levelized cost of electricity of 0.29 €/kWh, a carbon footprint of 0.20 kg CO₂ eq./kWh, and a cumulative energy demand of 4.88 MJ/kWh at a production cost of 0.67 €/cm² [109]. At multi-terawatt scale, bottom-up cost modeling indicates that recycling perovskite glass can reduce recycling costs by up to \$7 per square meter with scalability up to 4 TW, managing the 10 to 25 million tons of glass projected annually in Europe from 2035 to 2070 [110].

8. Research Gaps and Outlook

The current literature focuses disproportionately on single-junction optimization, even though certified single junctions have reached 33.1% while the best certified perovskite/silicon tandem this month reached 33.1%, demonstrating that the subcell that limits the tandem is not the one most papers are optimising. In tandem configurations, overall efficiency gains depend on the wide-bandgap top cell, which suffers from severe voltage deficits and phase instability. Although numerous chemical treatments claim to achieve halide homogenisation in wide-bandgap absorbers, evidence that this homogenisation suppresses phase segregation under real operating bias remains thin. Most evalua-

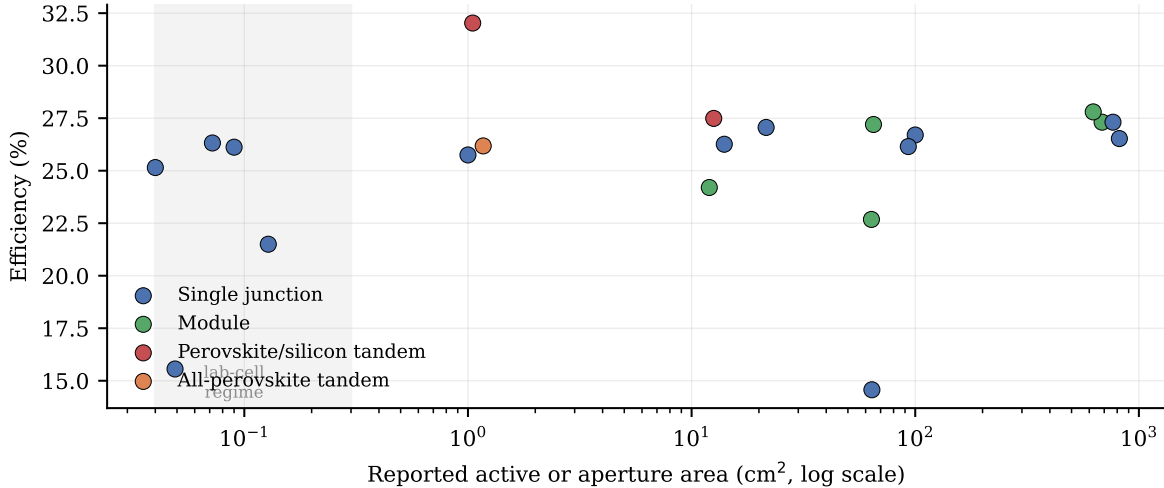


Figure 4: **Figure 4.** Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

tions rely on uncertified open-circuit photoluminescence or passive light soaking, ignoring how carrier extraction and electric fields drive iodide-bromide demixing into low-bandgap iodide clusters that pin the open-circuit voltage. Operando nanoscale photoluminescence spectroscopy coupled with depth-resolved Kelvin probe force microscopy under active current extraction at maximum power point conditions would determine whether halide mixing persists when carrier gradients and operational electric fields act simultaneously.

A second critical gap concerns buried-interface chemistry that survives thermal cycling rather than only improving initial efficiency. Interfacial modifiers such as self-assembled monolayers, phosphonic acids, and Lewis bases improve open-circuit voltage on unaged substrates, but these claims almost universally rest on small devices evaluated at room temperature. Under thermal cycling, mismatched thermal expansion coefficients between the perovskite absorber and rigid transport contacts induce severe biaxial lattice strain. Evidence conflicts on whether interfacial molecules relieve this stress or promote delamination through weak electrostatic bonding and thermal desorption. In situ grazing-incidence wide-angle X-ray scattering paired with operando transient photovoltage measurements during continuous temperature cycling would directly monitor dynamic lattice strain relaxation and interface recombination velocities, identifying chemical anchoring motifs that maintain electronic contact across thermal extremes.

Defect dynamics also remain poorly understood because ion-migration timescales are measured in the dark rather than under load. Current literature routinely extracts activation energies from dark impedance spectroscopy or capacitance-voltage sweeps. These dark measurements fail to capture the influence of photogenerated charge carriers on defect formation enthalpies and electrostatic screening. Under operating conditions, large photogenerated carrier densities screen internal electric fields, while chemical potential gradients accelerate mobile halide vacancy redistribution toward selective contacts. Measuring timescales in the dark obscures how mobile ions alter carrier recombination during active operation. Transient optoelectronic impedance spectroscopy and intensity-modulated photovoltage measurements collected under continuous illumination at the maximum power point would extract mobile ion drift velocities and interfacial accumulation rates under load, providing realistic kinetic

parameters for operational transport.

A pronounced disconnect persists between small-area interfacial engineering and scalable module manufacturing. Passivation that lifts Voc on a 0.1 cm² cell rarely survives slot-die coating at module area. Passivation recipes designed for rapid spin-coating and antisolvent quenching depend on non-equilibrium crystallization. In slot-die coating, slow drying kinetics and fluid-flow instabilities cause passivating molecules to aggregate into inhomogeneous clusters, leaving pinholes and unpassivated boundaries. The literature reflects a profound neglect of this transition: only 27 of 648 works addressed scale-up at all, and the largest reported aperture area was 818 cm², which is two to three orders of magnitude below a commercial module. High-speed operando photoluminescence imaging combined with in situ grazing-incidence small-angle X-ray scattering during slot-die printing would correlate ink drying dynamics and ligand distribution with spatial open-circuit voltage maps across large areas, establishing scalable processing boundaries for defect passivation.

Operational durability assessments are further obscured by the absence of a shared degradation protocol that would make lifetime claims comparable. Published studies alternate between unencapsulated shelf-life tests, open-circuit illumination, and room-temperature tracking without environmental control. Out of 169 closely read papers, only 4 reported a T80 lifetime, and the longest was 9800 h, which is short against a 25 year module warranty. These unstandardized, unlabelled stability evaluations disguise whether failure stems from intrinsic lattice degradation, interfacial chemical reaction, or contact corrosion. Adopting standardized consensus aging protocols that combine continuous maximum power point tracking under illumination with simultaneous thermal cycling and reverse-bias stress, accompanied by operando electrical impedance tracking, would isolate contact degradation from bulk absorber decay and enable rigorous lifetime comparisons.

The literature would become substantially more useful if research shifted away from reporting incremental efficiency gains on minute spin-coated cells toward mechanistic endurance under operational conditions. Priority must be given to operando evaluation of wide-bandgap subcells under bias, thermally stable buried interfaces that withstand cyclical strain, scalable deposition diagnostics that preserve passivation over large areas, and standardized accelerated aging metrics. Establishing these practices will bridge the divide between laboratory cell performance and commercially viable photovoltaic modules.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

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